Project Background

Artificial intelligence is a nation-defining capability that can lead to greater prosperity for all. From the application of LLM-generated video applications to deliver personalized education in rural communities to leveraging AI to provide real-time medical guidance during emergencies in areas where expert knowledge is limited, the possibilities are limitless.

However, this requires a more systematic approach which leverages extended technologies like Digital Public Infrastructure (DPI) to massify its reach and increate its utility value for the delivery of last-mile innovations for all segments of society in areas like health, financial inclusion, agriculture, and others. DPI is a digital network that enables countries to deliver economic opportunities and social services safely and efficiently to all residents. DPI can be compared to roads, rail lines, or any shared public utilities, which form physical networks connecting people and providing access to a huge range of goods and services. DPI serves as a combination of networked open technology standards built for public interest to enable governance and a community of innovative and competitive market players working to drive innovation. This is especially the case when its three foundational pillars of digital identification, payments, and data exchange are intentionally aligned for social impact.

As technologies become increasingly interconnected in building inclusive digital economies, there is an opportunity to leverage the convergence of AI and digital public good to accelerate the mass adoption of AI use cases for all segments of society. Combining emerging generative AI possibilities for nuanced local innovations and deploying on open-source DPI infrastructure has great potential to stimulate high impact and relevant local innovations, which may not be of immediate commercial relevance to revenue-oriented organizations. Integrating AI-based applications with digital public goods means more democratization and inclusion, which could create a fairer world where everyone can seamlessly enjoy personalized health services, run an ecommerce business from anywhere, produce digital art for sale and earn in foreign currency, write software, and access information to an extent never before possible. Moreover, the ease with which DPI-powered AI applications can be used and their cloud-based nature could be essential to narrowing the digital divide. For example, a student who gets a legal form of identity at birth that is linked via a data exchange system to clinics and pharmacies can receive better health care over a lifetime, and the service received will also become more personalized via AI knowledge.

Bangladesh provides an early illustration of how national vision can anchor AI-DPI integration, with its *Smart Bangladesh 2041* agenda positioning AI as a core enabler of service delivery, economic planning, and inclusion through practical deployments across government services, disease detection, education, and financial access, while explicitly linking AI adoption to DPI to minimize digital divides; India represents a more advanced stage of this evolution, demonstrating how embedding AI within open, interoperable DPI supported by open-source software, shared datasets, and common frameworks, can democratize AI at population scale, transforming DPI from static infrastructure into a dynamic, adaptive ecosystem that delivers personalized, user-centric services; this model is especially instructive for countries like Nigeria, where diverse linguistic, geographic, and socioeconomic conditions require AI systems integrated with public digital identity and trusted data layers to deliver localized, inclusive services such as native-language financial guidance for low-income populations; a concrete example of DPI-enabled AI in action is the Data in Climate Resilient Agriculture (DiCRA) platform developed with the United Nations Development Programme and the Government of Telangana, which uses open-source tools, remote sensing, and crowd-sourced data to generate farm-level climate resilience intelligence through a globally replicable, open data-and-analytics model; similarly, India's Bhashini national language platform demonstrates how AI and DPI together can overcome literacy and language barriers at scale by enabling multilingual speech recognition, translation, and voice-based access to government services, reinforcing the role of AI-enabled DPI as a foundational public good that drives inclusion, adaptability, and equitable access across societies.

Building a technical support ecosystem for the enablement of AI and DPI innovations

Solution Delivery Details

The project proposes to leverage the vast AI and DPI open-source platforms and democratized enablement platforms to massively scale AI/DPI application development through an interconnected network of ecosystem members: solution builders, problem owners, problem validators, solution users, and solution enablers. By creating an interconnected support network that connects diverse stakeholders, this project is poised to drive the scalable development of home-grown, nuanced, and locally relevant innovation built on uniquely combining the possibilities provided by AI and DPI in solving developmental challenges and enhancing public service delivery across communities in Nigeria.

The selected innovators from this grant were supported through an enabling technology stack that combines fundamental AI and DPI tools to build scalable and locally nuanced innovations. This was complemented by a network of local and international experts who provided hands-on support to the local innovators to embed best practice tools in AI and DPI for market readiness and solution deployment. The technology stack packaged into a cloud-based innovation sandbox provided access to an adapted foundational building block, GPU and models access to accelerate solution experimentation, and a host of other features that enables a mini "*Nigerian DPI stack*" that provides relevant interface to multiple open-source tools (e.g. DHIS2, PostgreSQL), APIs (e.g. identity) and apps (e.g. QR code, notification) as necessary.

The selected innovators had access to hands-on capacity building in AI+DPI application solutioning, solution customization, prototyping and enablement with DPI integration and end-to-end solution development support in high-impact areas like Health, Agriculture, Education, and Financial Inclusion.

Our solution delivery architecture approach was designed along three (3) standard DPI frameworks with robust building blocks that have been tried and tested to support multiple use cases that we had anticipated in the project.

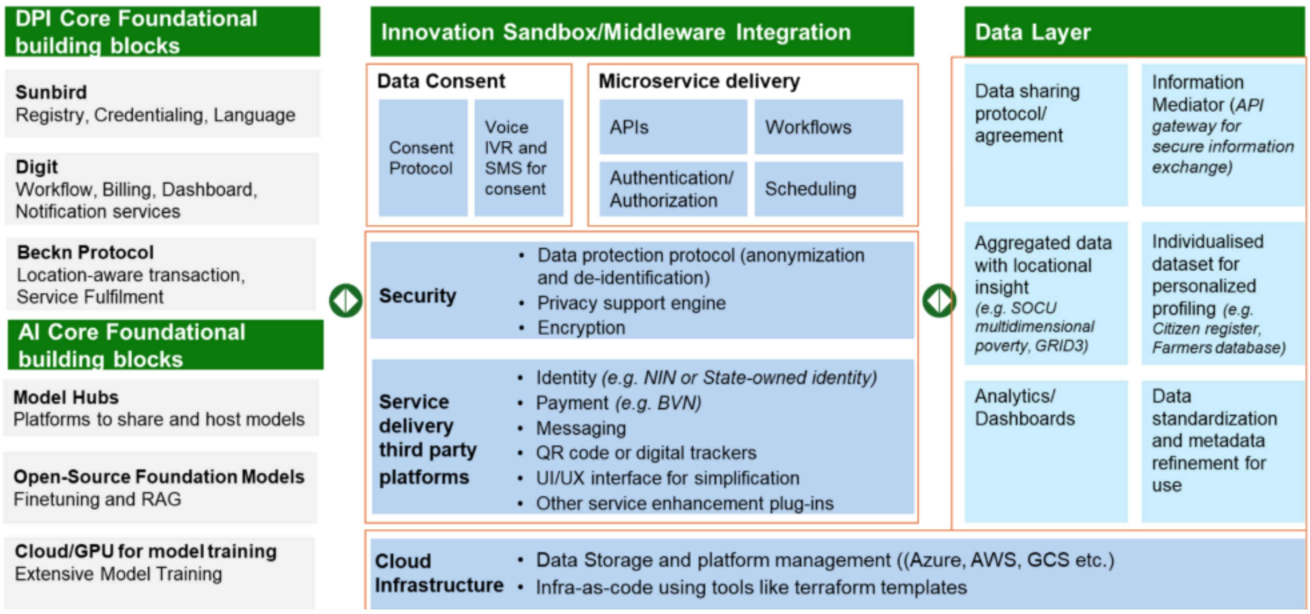


Figure 1: Proposed Integrated DPI + AI Solution Delivery Framework

Innovators Technical Support and Expertise Embedding:

In addition to the technology stack, the selected innovators had access to best-in-class solution development and deployment support. These are itemized below:

1. **Best practice solution guide and the documentation of related solutions from different parts of the world to accelerate development:** To establish a strong foundation for innovation, innovators were supported with a comprehensive archive of use cases demonstrating potential AI and DPI integrations within the focused domain. This use case documentation included existing and relevant DPI and AI solutions, providing a thorough understanding and systematic characterization of

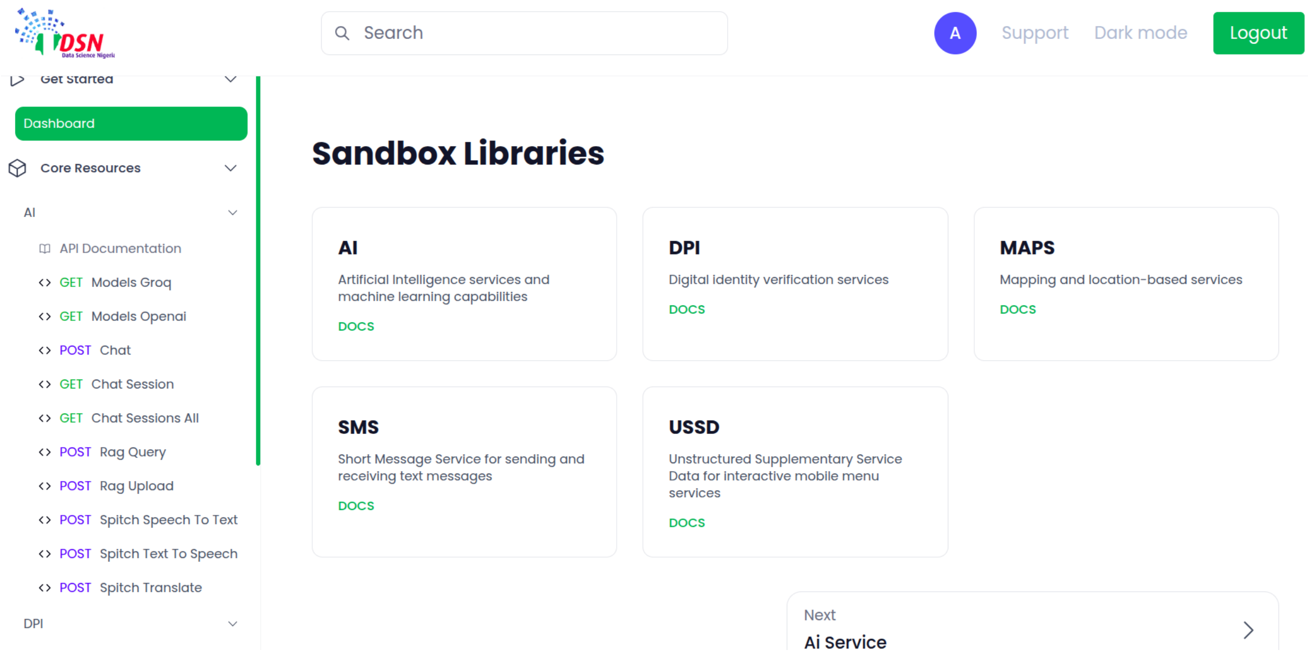
problems from the perspectives of both end-users and solution developers. This approach served as a valuable resource, guiding innovators as they explore and develop their own solutions.

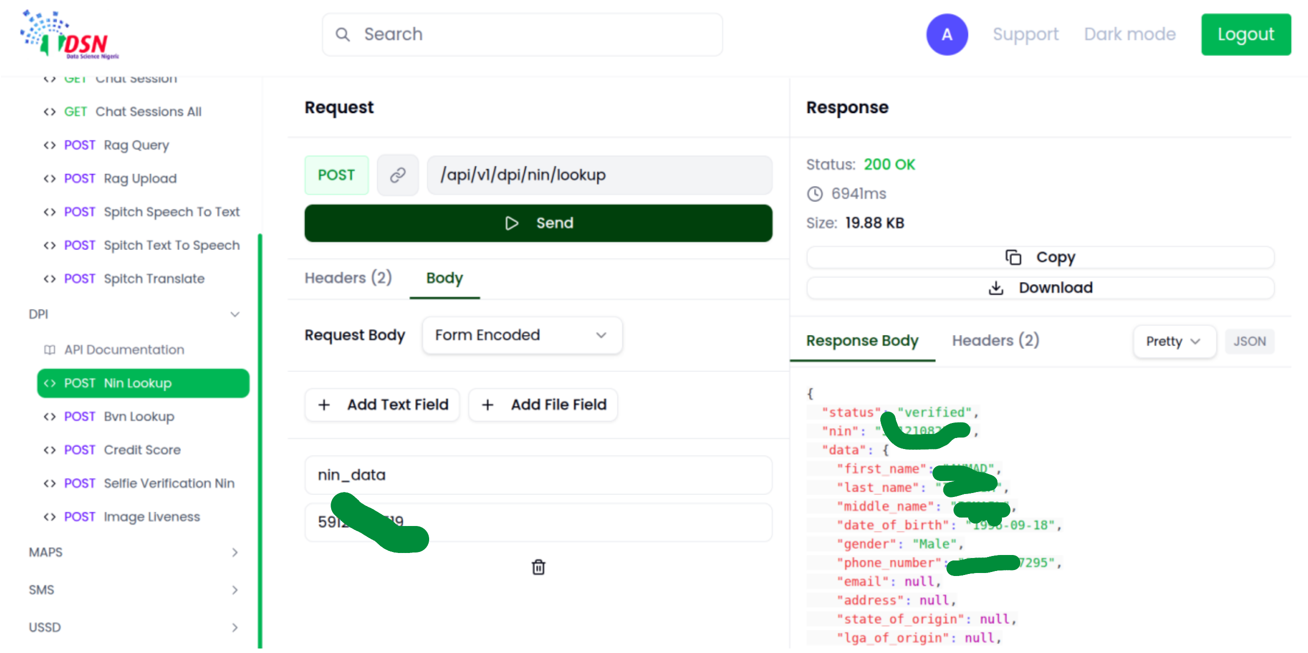
2. **Solution testing:** The Innovators had access to focus groups of testers that were available to test their solutions as they build for iterative refinement. The testing allowed them to identify potential flaws and areas for improvement before full-scale implementation, reducing the risk of failure. Innovators were able to gather valuable feedback from the team, leading to more refined and user-centric products. This iterative process fostered creativity and encouraged experimentation, enabling innovators to explore different approaches and find the most effective solutions. It was designed to ensure that the final product is robust, reliable, and ready for market adoption, enhancing the chances of success.
3. **Support on integration of open-source tools:** The platform also allowed access to open-source tools like DHIS2, Python, Apache etc. By leveraging open-source frameworks, AI innovators had the choice of experimenting with and customizing existing algorithms and models to fit their specific needs, fostering innovation and creativity.
4. **Access to solution development platform and model training GPUs**: The first layer of the platform provided the innovators with Cloud Infrastructure/GPUs for solution development and model training (e.g. Azure, AWS, Google Cloud Platform). This ensured an enhanced capability to handle large datasets and complex computations.
5. **Expert access for skill transfer and hands-on domain expert support (AI and DPI):** As part of the initial technical support, there was a 2-week intensive residency programme for the selected innovators who were new to the concept of DPI, and in some cases AI. Experts were be available to take sessions on product design, ideation, AI, DPI and the integration of AI into DPI. This comprehensive programme ensure that innovators had access to the necessary skills and knowledge to effectively utilize DPI infrastructure and integrate AI models, driving the usability and effectiveness of their solutions.

The AI-DPI Sandbox

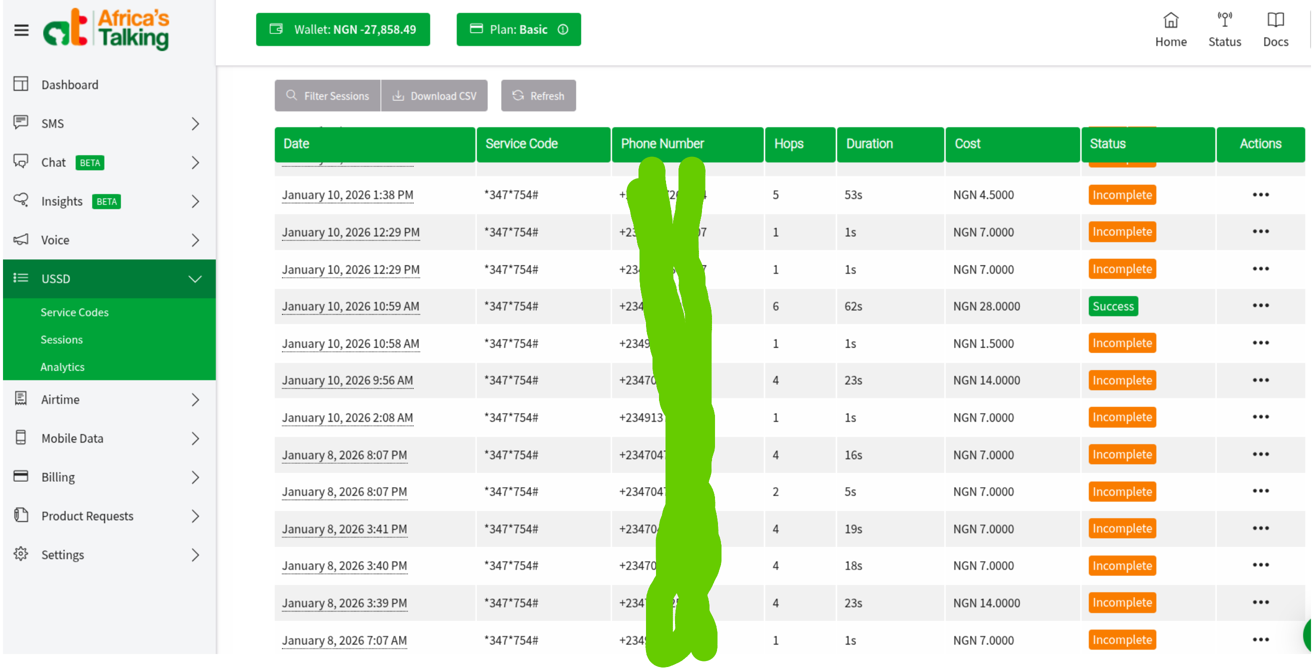
The AI-DPI Sandbox was designed to accelerate responsible, inclusive, and scalable adoption of Artificial Intelligence (AI) and Nigeria's Digital Public Infrastructure (DPI) by early-stage startups and innovators. The platform responded to a critical ecosystem challenge: while AI and DPI platforms are increasingly available, startups often face high integration costs, fragmented vendor ecosystems, regulatory complexity, and limited technical capacity to adopt these systems effectively.

The fully managed, secure, API-based sandbox environment unified access to AI services, Nigerian DPI platforms, and essential network services (SMS, USSD, IVR, voice). Complementing the sandbox, the project delivered intensive, hands-on technical support to participating startups, ensuring not only access to infrastructure but also the practical capability to integrate, test, iterate, and deploy solutions aligned with regulatory, security, and ethical requirements.





The sandbox was intentionally designed as a market-enablement and capacity-building intervention rather than a standalone technical product. By acting as an intermediary layer between startups and multiple vendors, including AI model providers, telecommunications platforms, and DPI service provides, the project significantly reduced time-to-market, lowered cost barriers, and de-risked early-stage experimentation. This allowed startups to focus on product innovation, user experience, and impact delivery while operating within a controlled and compliant environment.



AI-DPI Sandbox Web Application Technology Stack

Technology	Purpose
Sandbox Web App	
Frontend	React, Tailwind
Backend	NodeJS/Next, FastAPI
Authentication and Authorization It provides a secure, scalable, and interoperable identity management system for users and applications within the sandbox.	
Oauth2	Its simple and can be implemented into Python/ FastAPI directly. **Features**: User federation, single sign-on (SSO), multi-factor authentication (MFA), role-based access control (RBAC), and API key management
Monitoring & Evaluation To provide comprehensive observability into the sandbox infrastructure and startup applications, enabling proactive issue detection, performance optimization, and data-driven decision-making.	
Prometheus	For collecting time-series metrics from all sandbox components (databases, LLM services, authentication) and potentially from startup applications (if they expose Prometheus endpoints)
Node Exporters/Service Monitors	Standardized exporters for common services and custom exporters for specific sandbox components.
Grafana	For creating interactive dashboards to visualize metrics and setting up alerts based on predefined thresholds

Jaeger/OpenTelemetry	For distributed tracing to understand request flows across microservices, aiding in debugging complex issues.
Caching and (Redis and related) To improve application performance, reduce database load, and enhance user experience by storing frequently accessed data in memory.	
Redis	Session Management API Response Caching: To speed up frequently requested data. Full-Page Caching: For static or semi-static content. Message Broker (Pub/Sub): For real-time event-driven communication between microservices Rate Limiting: To protect APIs from abuse.
ElasticSearch To provide powerful full-text search capabilities and analytical insights for large volumes of data, such as logs, health records, and product catalogs.	
Elasticsearch Cluster	A distributed, RESTful search and analytics engine
Kibana	For visualizing data stored in Elasticsearch and building dashboards.
Logstash/Beats	For data ingestion from various sources into Elasticsearch (can be integrated with the Monitoring & Logging stack).
Deployment	
Environments	Staging, and Production environments with distinct configurations and resource allocations.
Docker, helmfile	Deployment and microservice management.
Azure	Cloud infrastructure provision.

Results and Achievements

Over the course of the project, the AI-DPI Sandbox successfully enabled participating startups to integrate AI and DPI services that would otherwise have been cost-prohibitive or technically complex. Startups were able to prototype, test, and refine products that leverage identity verification, multilingual AI, and multi-channel communication within significantly reduced development timelines.

The project demonstrated that a managed sandbox model, coupled with sustained technical support, can materially improve startup readiness, reduce integration risk, and accelerate time-to-market. Importantly, it also validated the value of DPI as a practical enabler of inclusive digital services when access barriers are removed.

DSN Residency Program Report

Phase 1: Orientation and Prototyping (Weeks 1 to 4)

In August, our focus was on laying a strong foundation for success. We began by welcoming the startups and immersing them in the program vision. This was not just an introduction; it was about creating clarity and confidence. We worked closely with each team to understand their problem statements, validate assumptions, and analyze available data for actionable insights. From there, we guided them through framing solutions and mapping how artificial intelligence and digital public infrastructure could create real public value. By the end of this phase, every startup had a clearly defined use case, a chosen technology stack, and a roadmap for their minimum viable product. This stage transformed raw ideas into structured, executable plans, ensuring teams were aligned and ready for the build phase.

Phase 2: Iteration and MVP (Weeks 5 to 8)

September was the engine room of the program, the point where ideas became tangible products. Startups moved from planning to execution, focusing on building and refining their minimum viable products. We supported them through feature prioritization, sprint planning, and test-driven development. Accessibility and inclusion were key themes, ensuring solutions could serve diverse user bases. Compliance was another critical area, especially for health and finance-focused products, so we embedded legal and regulatory checks early. Internal usability testing helped identify pain points, and teams iterated quickly to address them. By the end of this phase, every startup had a functional minimum viable product aligned with its product requirements, ready for real-world validation.

Phase 3: Testing, Deployment and Growth (Weeks 9 to 12 and Early November)

October and extending into early November was all about validation, deployment, and preparing for scale. We initiated live user testing, collected structured feedback, and iterated rapidly to improve usability. Technical rigor was a priority: staging environments, load and security testing, and deployment readiness checks ensured stability and resilience. Soft launches allowed us to monitor real-world performance, track metrics, and fix critical bugs before full release. Beyond the technical work, we prepared startups for investor engagement. Pitch rehearsals, demo materials, and strategic handovers positioned them for growth beyond the residency. This phase was not just about launching products; it was about proving impact, demonstrating scalability, and setting the stage for long-term success.

In summary: Over twelve weeks and into early November, we took startups through a deliberate journey: clarity in Phase One, execution in Phase Two, and validation plus scale in Phase Three. Each step was designed to reduce risk, accelerate learning, and maximize impact. Today, these teams are not just demo-ready; they are market-ready, with tested solutions, validated impact, and a clear path forward. We are particular excited about the Pilot phase and can wait to see the impact these startups will make.

Here is a list of the startups and the solution their pilot-ready solutions:

Startup	**Name of Solution**
MyItura	MediLoan
Clafiya	Clafiya

UHC Tech	EasyCover
Fertitude	Fertitude
Flolog Pharma	MomCare
Alajo	Alajo App
XChangeBox	Kidashi
Evet Africa	Agrocist
Eight Medical	Maternal Lite

Lessons Learned

Readiness and proximity of technical support as a catalyst for adoption: A key lesson from the AI-DPI Sandbox and Technical Support Project is that the availability, responsiveness, and proximity of technical support significantly influence the pace and depth of technology adoption by startups. Many participating teams had strong product ideas but varying levels of experience with complex integrations involving AI services, telecommunications infrastructure, and national digital public infrastructure. By positioning technical support as an embedded, ongoing function rather than an ad hoc troubleshooting service, the project reduced friction at critical points in the development cycle. Proactive engagement through regular check-ins and guided integration helped startups resolve issues early, avoid architectural missteps, and maintain development momentum.

Sandbox environments as enablers of safe, iterative innovation: The project also demonstrated the strategic value of sandbox environments in fostering innovation, particularly in high-risk or highly regulated domains. By providing a controlled, isolated environment that mirrored real-world infrastructure, the sandbox allowed startups to test ideas, iterate rapidly, and explore multiple service combinations without the fear of disrupting live systems or incurring prohibitive costs from direct vendor usage. This freedom encouraged experimentation across AI models, communication channels, and DPI integrations, often leading startups to expand beyond their initial product assumptions. The sandbox effectively lowered the cost of failure, making iteration a natural part of the development process rather than a risk to be avoided. As a result, startups were able to refine their products more confidently and responsibly, emerging with solutions that were both technically sound and better aligned with real-world deployment conditions.